

ACRYLIC FIBERS

Definition

Is defined by TPPIA as a manufactured fibre in which the fibre forming substance is any long-chain synthetic polymer composed of at least 85% by weight of acrylonitrile units.

Production of Acrylic Fibres

Acrylic fibres are polymers formed by additional polymerisation.

- Acrylonitrile or vinyl cyanide is required. The first acrylic fibre was orlon in 1950 by Du Pont company and since then many changes have been made.
- Acrylonitrile is added to water containing a proxy catalyst. As the polymer precipitates, it is filtered or washed, dried and dissolved in a suitable solvent.
 - Typical co-monomers are vinyl acetate or methyl acrylate. DuPont created the first acrylic fibers in 1941 and trademarked them under the name Orlon.

- The fibres may be wet spun into a coagulating bath or they may be dry spun with the solvent evaporated off in hot air.

Both dry and wet spinning processes are used to make acrylic fibres. In the dry-spinning process the spinning mass emerges from the spinneret into a spinning chamber, into which warm air is blown. In wet spinning the spin mass is pressed into a so-called coagulating bath. The largest proportion of the production is made and used as crimped staple fibre.

- The solvent usually used for spinning poly-acrylonitrile is dimethyl formamide.

- Examples Acrilan, orlon courtell.....

Basic Principles of Acrylic Fiber Production

Acrylic fibers are produced from **acrylonitrile**, a **petrochemical**. The acrylonitrile is usually combined with small amounts of other chemicals to improve the ability of the resulting fiber to absorb dyes. Some acrylic fibers are dry spun and others are wet spun. Acrylic fibers are used in staple or tow form.

Acrylic fibers are modified to give special properties best suited for particular end-uses. They are unique among synthetic fibers because they have an uneven surface, even when extruded from a round-hole spinneret.

Acrylic Fiber Characteristics/properties

- Outstanding wickability & quick drying to move moisture from body surface.
- Flexible aesthetics for wool-like, cotton-like, or blended appearance.
- Easily washed, **retains shape**.
- Resistant to moths, oil, and chemicals.
- Dyeable to bright shades with excellent fastness.
- Superior resistance to sunlight degradation.

Use and Care of Acrylics

Major acrylic fiber uses are as follows;

1. **Apparel:** Sweaters, socks, fleece wear, circular knit apparel, sportswear and children's wear.

2. **Home Furnishings:** Blankets, area rugs, upholstery, pile; luggage, awnings, outdoor furniture
3. **Other Uses:** Craft yarns, sail cover cloth, wipe cloths
4. **Industrial Uses:** Asbestos replacement; concrete and stucco reinforcement.

General Acrylic Fiber Care Tips

Wash delicate items by hand in warm water. Static electricity may be reduced by using a fabric softener in every third or fourth washing. Gently squeeze out water, smooth or shake out garment and let dry on a non-rust hanger. (Sweaters, however, should be dried flat.)

When machine washing, use warm water and add a fabric softener during the final rinse cycle.

Machine dry at a low temperature setting. Remove garments from dryer as soon as tumbling cycle is completed.

If ironing is required, use a moderately warm iron.